

Truth Values

The truth value of a mathematical statement is not determined in isolation, but always relative to a universe of discourse or context, in which the statement is interpreted. For example, consider the statement

$$\forall x, y \exists z ((x \neq y) \implies (x < z < y))$$

which may be read as

"For any two distinct numbers x, y , there exists a number z strictly between them."

Note that if the statement is interpreted in the context of natural numbers, it is false because there is no natural number between 1 and 2. But if the statement is interpreted in the context of rational numbers, it is true because for any two rational numbers x, y , the number $z = \frac{x+y}{2}$ is a rational number between them.

- **Definition 1**

Let P and Q be mathematical statements interpreted in some context u , the truth value of $P \wedge Q$ in such a context is fully determined by the following truth table:

P	Q	$P \wedge Q$
1	1	1
1	0	0
0	1	0
0	0	0

- **Remark 1**

The truth table for conjunction can be understood in an even more compact way as follows:

- **Example 1.1**

If $\|P\|$ is the truth value of P and $\|Q\|$ is the truth value of Q , then

$$\|P \wedge Q\| = \min \{ \|P\|, \|Q\| \}$$

This algebraic perspective is useful for replacing structural properties of logical connectives.

- **Definition 2**

Let P and Q be mathematical statements interpreted in some context u , the truth value of $P \vee Q$ in such a context is fully determined by the following truth table:

P	Q	$P \vee Q$
1	1	1
1	0	1
0	1	1
0	0	0

- **Remark 2**

The truth table for disjunction can be understood in an even more compact way as follows:

- **Example 2.1**

If $\|P\|$ is the truth value of P and $\|Q\|$ is the truth value of Q , then

$$\|P \vee Q\| = \max \{ \|P\|, \|Q\| \}$$

- **Definition 3**

Let P and Q be mathematical statements interpreted in some context u , the truth value of $P \Rightarrow Q$ in such a context is fully determined by the following truth table:

P	Q	$P \Rightarrow Q$
1	1	1
1	0	0
0	1	1
0	0	1

- **Remark 3**

In algebraic form, the truth table can be understood in an even more compact way as follows:

- **Example 3.1**

If $\|P\|$ is the truth value of P and $\|Q\|$ is the truth value of Q , then

$$\|P \Rightarrow Q\| = \begin{cases} \|Q\| & \text{if } \|P\| = 1 \\ 1 & \text{otherwise} \end{cases}$$

or in another form:

$$\|P \Rightarrow Q\| = \max \{1 - \|P\|, \|Q\|\}$$

- **Definition 4**

Let P and Q be mathematical statements interpreted in some context u , the truth value of $P \Leftrightarrow Q$ in such a context is fully determined by the following truth table:

P	Q	$P \Leftrightarrow Q$
1	1	1
1	0	0
0	1	0
0	0	1

- **Remark 4**

In algebraic form, the truth table can be understood in an even more compact way as follows:

- **Example 4.1**

If $\|P\|$ is the truth value of P and $\|Q\|$ is the truth value of Q , then

$$\|P \Leftrightarrow Q\| = \min (1 - \|P\| + \|Q\|, 1 - \|Q\| + \|P\|)$$

- **Definition 5**

Let P be a mathematical statement interpreted in some context u , the truth value of $\neg P$ in such a context is fully determined by the following truth table:

P	$\neg P$
1	0
0	1

- **Remark 5**

In algebraic form, the truth table can be understood in an even more compact way as follows:

- **Example 5.1**

If $\|P\|$ is the truth value of P , then

$$\|\neg P\| = 1 - \|P\|$$

We now turn to the question of how truth values are assigned to statements involving quantifiers.

- **Definition 6**

A universally quantified statement of the form $\forall xP(x)$ is true if and only if $P(x)$ is true for all possible values of x . Note that the universally quantified statement $\forall xP(x)$ is false if $P(x)$ is false for some value of x . An existentially quantified statement of the form $\exists xP(x)$ is true if $P(x)$ is true for some value of x . Note that the existentially quantified statement $\exists xP(x)$ is false if $P(x)$ is false for all possible values of x .

- **Remark 6**

In algebraic form, the truth values of the qu/antifiers $\forall xP(x)$ and $\exists xP(x)$ in a context u are:

- **Example 6.1**

$$\|\forall xP(x)\| = \min_{x \text{ takes a value in } u} \{\|P(x)\|\}$$

$$\|\exists xP(x)\| = \max_{x \text{ takes a value in } u} \{\|P(x)\|\}$$

- **Definition 7**

We say that two mathematical statements P and Q are logically equivalent, denoted by $P \equiv Q$, if for any context u we obtain $\|P\| = \|Q\|$.

- **Example 7.1**

Let P and Q be two mathematical statements, then $\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$.

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$(\neg P \vee \neg Q)$
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

Therefore, $\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$.

- **Example 7.2**

Let P and Q be two mathematical statements, then $\neg(P \implies Q) \equiv P \wedge \neg Q$.

P	Q	$P \implies Q$	$\neg(P \implies Q)$	$\neg Q$	$(P \wedge \neg Q)$
1	1	1	0	0	0
1	0	0	1	1	1
0	1	1	0	0	0
0	0	1	0	1	0

Therefore, $\neg(P \implies Q) \equiv P \wedge \neg Q$.

- **Example 7.3**

Determine if $\forall xP(x) \equiv \neg\exists x(\neg P(x))$.

- **Solution 7.3.1**

We consider two possible cases for the truth value of $\forall xP(x)$ in any given context u :

Case 1: If $\forall xP(x)$ is true, then $\|\forall xP(x)\| = 1$. This implies that $\|P(x)\| = 1$ for all x in u . Consequently, the truth value of its negation is $\|\neg P(x)\| = 1 - \|P(x)\| = 0$ for all x in u . Taking the maximum over all x , we get:

$$\|\exists x(\neg P(x))\| = \max_x \{\|\neg P(x)\|\} = 0$$

Thus, its negation evaluates to:

$$\|\neg\exists x (\neg P(x))\| = 1 - 0 = 1$$

Case 2: If $\forall x P(x)$ is false, then $\|\forall x P(x)\| = 0$. This implies that there exists at least one element x_0 in u such that $\|P(x_0)\| = 0$. For this specific element, we have $\|\neg P(x_0)\| = 1 - \|P(x_0)\| = 1$. Since there is at least one element where $\neg P(x)$ is true, taking the maximum yields:

$$\|\exists x (\neg P(x))\| = \max_x \{\|\neg P(x)\|\} = 1$$

Thus, its negation evaluates to:

$$\|\neg\exists x (\neg P(x))\| = 1 - 1 = 0$$

In both cases, we obtain:

$$\|\forall x P(x)\| = \|\neg\exists x (\neg P(x))\|$$

By Definition 2.7, we conclude that $\forall x P(x) \equiv \neg\exists x (\neg P(x))$.